Comprehensive Practice: Quadratic Sequences

Mathematics Department

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Student Information

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16+8+4 = 28 → 28 ÷ 2 = 14 → 35÷2 = 19.5

25+10+4 = 39 Prone to sweep! ☺

Question 1

The nth term of this sequence is:

$$T_n = \frac{n^2 + 2n + 4}{2}$$

1+2=3 3+4=7

7÷2 = 3.5

Task A: Complete the Table

n	1	2	3	4	5	6
T_n	3.5	6	9.5	14	19.5	

9+6+4=19

19÷2 = 9.5

Task B: Prediction and Verification

– The pattern starts with 3.5, and the difference between each term increases by 1.5 as you progress. | The pattern starts at 3.5. As the sequence progresses, the difference between current term & next term increases by 1 each time.

i. Predict the **10th term** (T_{10}) using the pattern of differences: T_n = 62

ii. Show the calculation using the formula for $n = 10$:

10×10 = 100 → 100 + 20 → 120 + 4 → $\frac{124}{2}$ = 62

36+18+1 = 55

25+15+1 = 41

9+9+1 = 19

16+12+1 = 29

1+3+1 = 5

4+6+1 = 11

+1 36 + 19 = 55

Question 2

The nth term of this sequence is:

$$T_n = n^2 + 3n + 1$$

Task A: Complete the Table

+1 +8 +10 +12 +14

n	1	2	3	4	5	6
T_n	5	11	19	29	41	55

Task B: Prediction and Verification

This is a quadratic pattern that starts with 5 and as the pattern progresses, the difference between the current & next number will increase by 2

i. Predict the **10th term** (T_{10}) using the pattern of differences: 131

16 18 20 22

4×2=8

ii. Show the calculation using the formula for $n = 10$:

n^2+3n+1 $100+30+1=131$

Question 3

The nth term of this sequence is:

3×3 = 9-2=18 -> 18-5= 13

$$T_n = 2n^2 - 5$$

27 6 12 7 5 67

Task A: Complete the Table +6 +10 +14 +18 32-5 +22

n	1	2	3	4	5	6
T_n	-3	3	13	27	45	67

Task B: Prediction and Verification Quadratic pattern that starts with -3 and as the pattern progresses, the difference between the current & next value increases by 4

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 4

The nth term of this sequence is:

$$T_n = \frac{n^2 + n}{2}$$

Task A: Complete the Table 2 3 4 5 6

n	1	2	3	4	5	6
T_n	1	3	6	10	15	21

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 5

The nth term of this sequence is:

$$T_n = n^2 - 2n + 10$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 6

The nth term of this sequence is:

$$T_n = 3n^2 + 2$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 7

The nth term of this sequence is:

$$T_n = n^2 + 4n - 3$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 8

The nth term of this sequence is:

$$T_n = \frac{n^2 + 10}{2}$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 9

The nth term of this sequence is:

$$T_n = 2n^2 + n$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 10

The nth term of this sequence is:

$$T_n = n^2 - n + 5$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 11

The nth term of this sequence is:

$$T_n = (n+1)^2$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 12

The nth term of this sequence is:

$$T_n = 4n^2 - 10$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 13

The nth term of this sequence is:

$$T_n = \frac{2n^2 + 4}{2}$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 14

The nth term of this sequence is:

$$T_n = n^2 + 5n$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 15

The nth term of this sequence is:

$$T_n = 0.5n^2 + 4$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 16

The nth term of this sequence is:

$$T_n = n^2 + 11$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 17

The nth term of this sequence is:

$$T_n = 3n^2 - 2n$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 18

The nth term of this sequence is:

$$T_n = 100 - n^2$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 19

The nth term of this sequence is:

$$T_n = n^2 + 2n + 5$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 20

The nth term of this sequence is:

$$T_n = 2n^2 - 1$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$:

Question 21

The nth term of this sequence is:

$$T_n = n^2 + 6$$

Task A: Complete the Table

n	1	2	3	4	5	6
T_n						

Task B: Prediction and Verification

i. Predict the **10th term** (T_{10}) using the pattern of differences: _____

ii. Show the calculation using the formula for $n = 10$: